

Iteration 1: Data Loading, EDA, Preprocessing & Baseline MLP Model

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score, precision_score, recall_score
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import GridSearchCV
import warnings
from sklearn.exceptions import ConvergenceWarning

# Set global visualization style
sns.set_theme(style="whitegrid")

# 1. Load the dataset
# Ensure 'water_potability.csv' is in your working directory
df = pd.read_csv("water_potability.csv")
print("Dataset Shape:", df.shape)
df.head()
```

Dataset Shape: (3276, 10)

Out[1]:

	ph	Hardness	Solids	Chloramines	Sulfate	Conductivity	Organic_carbon	Trihalomethanes	Turbidity	Potability
0	NaN	204.890455	20791.318981	7.300212	368.516441	564.308654	10.379783	86.990970	2.963135	0
1	3.716080	129.422921	18630.057858	6.635246	NaN	592.885359	15.180013	56.329076	4.500656	0
2	8.099124	224.236259	19909.541732	9.275884	NaN	418.606213	16.868637	66.420093	3.055934	0
3	8.316766	214.373394	22018.417441	8.059332	356.886136	363.266516	18.436524	100.341674	4.628771	0
4	9.092223	181.101509	17978.986339	6.546600	310.135738	398.410813	11.558279	31.997993	4.075075	0

```
In [2]: # 2. Exploratory Data Analysis & Class Balance
print("Missing values per column prior to imputation:")
print(df.isnull().sum())

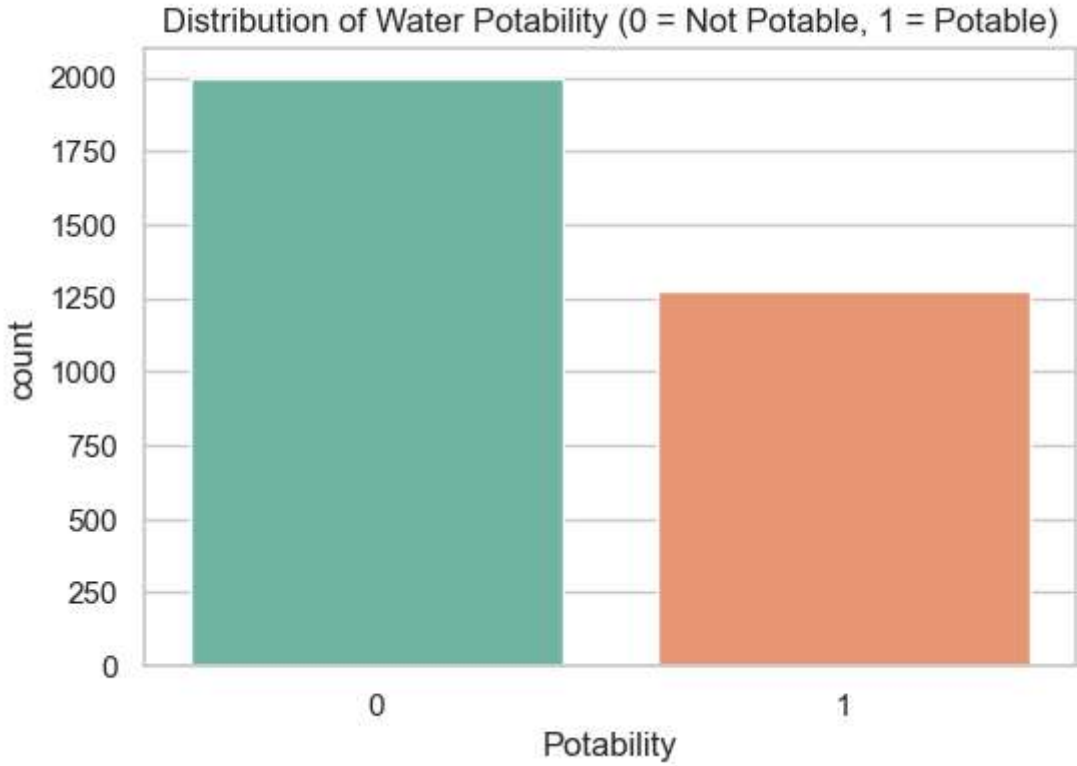
plt.figure(figsize=(6, 4))
sns.countplot(x='Potability', data=df, hue='Potability', palette='Set2', legend=False)
plt.title('Distribution of Water Potability (0 = Not Potable, 1 = Potable)')
plt.show()

print("\nClass distribution ratios:")
print(df['Potability'].value_counts(normalize=True))
```

Missing values per column prior to imputation:

ph	491
Hardness	0
Solids	0
Chloramines	0
Sulfate	781
Conductivity	0
Organic_carbon	0
Trihalomethanes	162
Turbidity	0
Potability	0

dtype: int64



Class distribution ratios:
Potability
0 0.60989
1 0.39011
Name: proportion, dtype: float64

```
In [3]: # 3. Safe Missing Value Imputation & Train-Test Split
df_clean = df.copy()
for col in df_clean.columns:
    if df_clean[col].isnull().sum() > 0:
        median_val = df_clean[col].median()
        df_clean[col] = df_clean[col].fillna(median_val)

print("Missing values after robust median imputation:", df_clean.isnull().sum().sum())

X = df_clean.drop('Potability', axis=1)
y = df_clean['Potability']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
print(f"Train set size: {X_train.shape[0]}, Test set size: {X_test.shape[0]}")
```

Missing values after robust median imputation: 0
Train set size: 2620, Test set size: 656

```
In [4]: # 4. Feature Scaling (Standardization)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
In [5]: # 5. Baseline MLP Model Training & Evaluation
mlp_baseline = MLPClassifier(
    hidden_layer_sizes=(64,),
    activation='relu',
    solver='adam',
    max_iter=500,
    random_state=42
)

print("Training baseline MLP model...")
mlp_baseline.fit(X_train_scaled, y_train)

y_pred = mlp_baseline.predict(X_test_scaled)

# Save scores for final comparison
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)

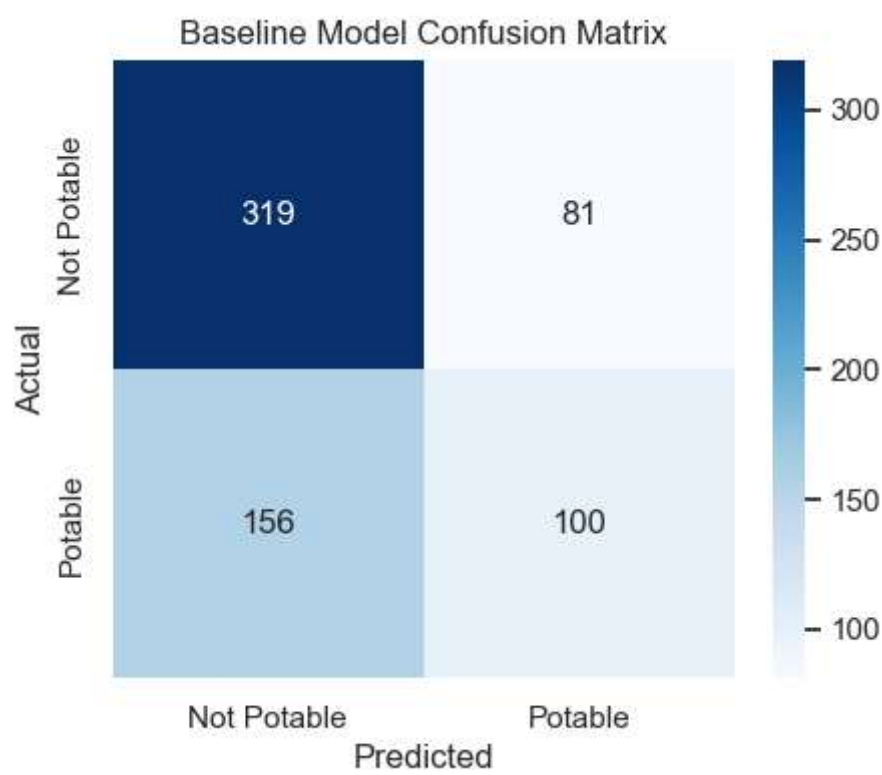
print("\n===== BASELINE PERFORMANCE =====")
print(f"Accuracy: {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall: {recall:.4f}")
print("=====")

plt.figure(figsize=(5, 4))
sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d', cmap='Blues',
    xticklabels=['Not Potable', 'Potable'], yticklabels=['Not Potable', 'Potable'])
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.title('Baseline Model Confusion Matrix')
plt.show()
```

Training baseline MLP model...

===== BASELINE PERFORMANCE =====
Accuracy: 0.6387
Precision: 0.5525
Recall: 0.3906
=====

c:\Users\xyz\Github\BIAI\.venv\Lib\site-packages\sklearn\neural_network_multilayer_perceptron.py:785: ConvergenceWarning: Stochastic Optimizer: Maximum iterations (500) reached and the optimization hasn't converged yet.
warnings.warn(



Iteration 2: Feature Space Reduction (Correlation & Feature Importance Analysis)

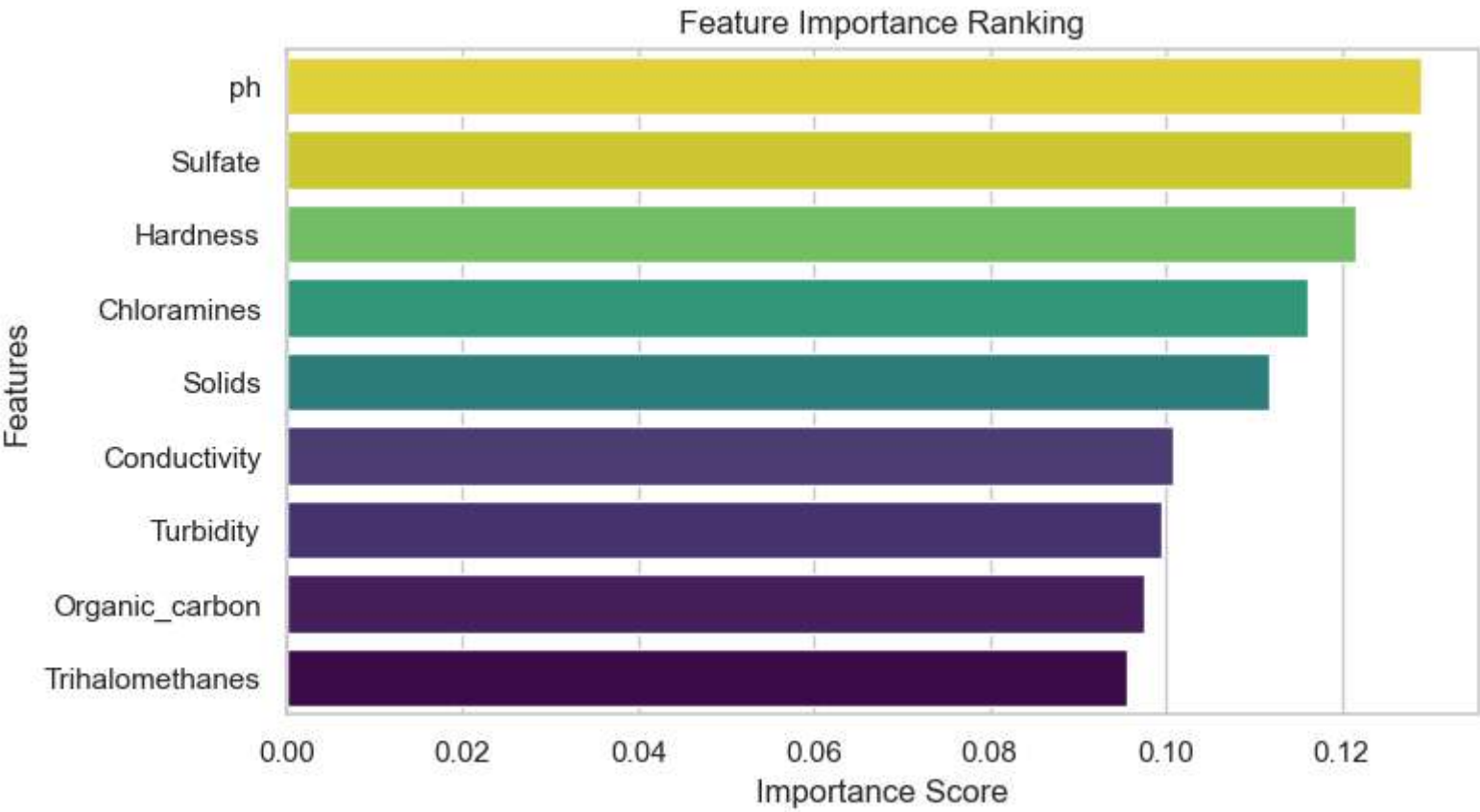
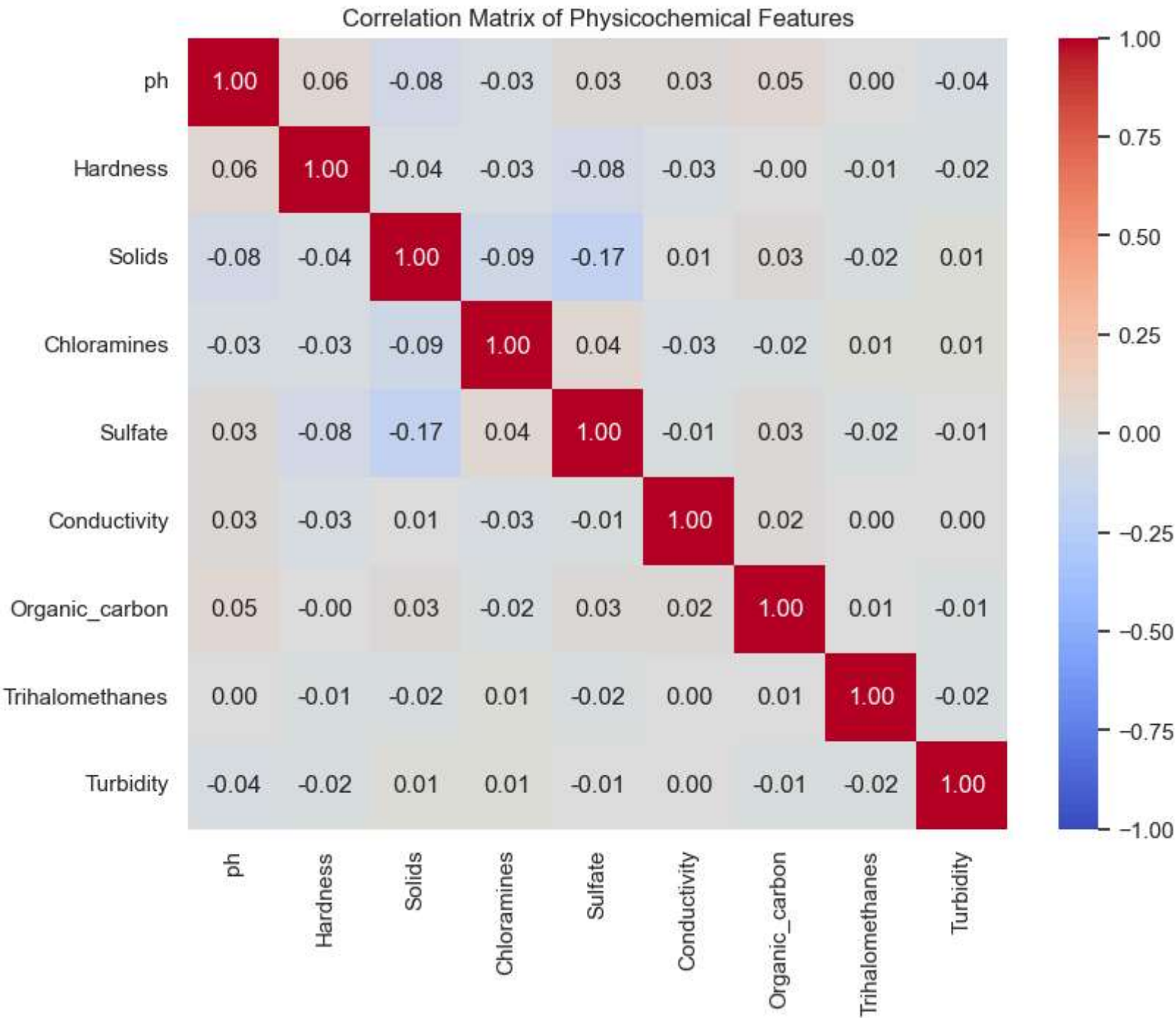
```
In [6]: # 1. Correlation Matrix Analysis
plt.figure(figsize=(9, 7))
corr_matrix = pd.DataFrame(X_train_scaled, columns=X.columns).corr()
sns.heatmap(corr_matrix, annot=True, fmt=".2f", cmap="coolwarm", vmin=-1, vmax=1)
plt.title("Correlation Matrix of Physicochemical Features")
plt.show()

# 2. Feature Importance Ranking via Random Forest Proxy
rf = RandomForestClassifier(random_state=42)
rf.fit(X_train_scaled, y_train)

feature_imp_df = pd.DataFrame({
    'Feature': X.columns,
    'Importance': rf.feature_importances_
}).sort_values(by='Importance', ascending=False)

plt.figure(figsize=(8, 4.5))
sns.barplot(hue='Importance', x='Importance', y='Feature', data=feature_imp_df, palette='viridis', legend=False)
plt.title('Feature Importance Ranking')
plt.xlabel('Importance Score')
plt.ylabel('Features')
plt.show()

print(feature_imp_df.to_string(index=False))
```



Feature	Importance
ph	0.129132
Sulfate	0.127858
Hardness	0.121633
Chloramines	0.116048
Solids	0.111863
Conductivity	0.100752
Turbidity	0.099522
Organic_carbon	0.097521
Trihalomethanes	0.095672

```
In [7]: # 3. Drop Lowest Impact Features & Re-train
features_to_drop = ['Organic_carbon', 'Trihalomethanes']
print(f"Dropping lowest impact features: {features_to_drop}")

feature_names = list(X.columns)
keep_indices = [feature_names.index(col) for col in feature_names if col not in features_to_drop]

X_train_reduced = X_train_scaled[:, keep_indices]
X_test_reduced = X_test_scaled[:, keep_indices]
```

```
mlp_reduced = MLPClassifier(
    hidden_layer_sizes=(64,),
    activation='relu',
    solver='adam',
    max_iter=500,
    random_state=42
)
mlp_reduced.fit(X_train_reduced, y_train)

y_pred_reduced = mlp_reduced.predict(X_test_reduced)
accuracy_red = accuracy_score(y_test, y_pred_reduced)
precision_red = precision_score(y_test, y_pred_reduced)
recall_red = recall_score(y_test, y_pred_reduced)

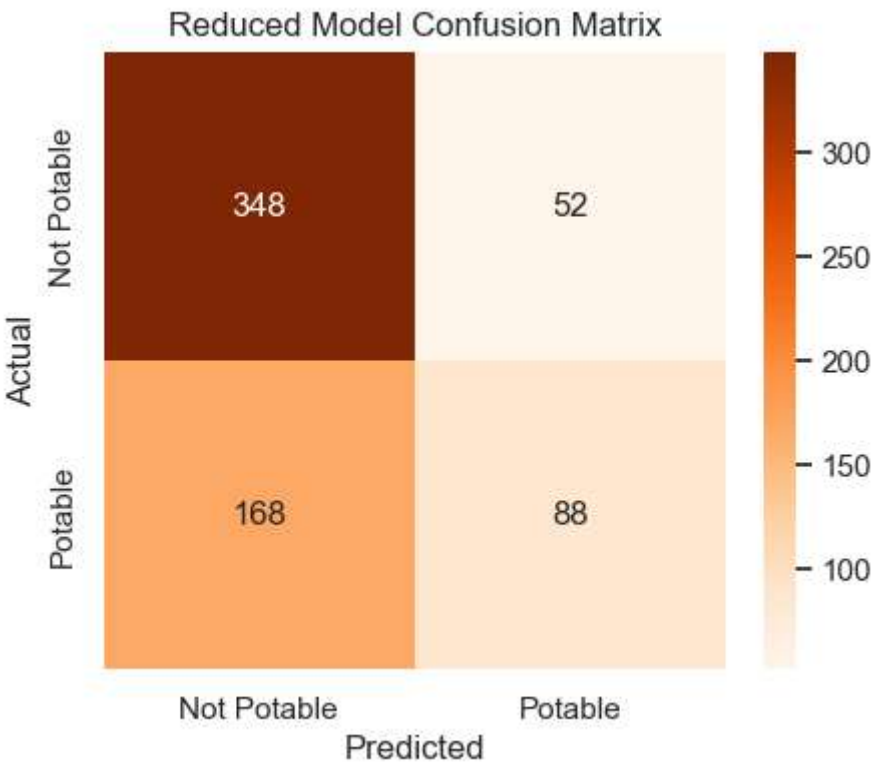
print("\n===== REDUCED MODEL PERFORMANCE =====")
print(f"Accuracy:  {accuracy_red:.4f} (Baseline: {accuracy:.4f})")
print(f"Precision: {precision_red:.4f} (Baseline: {precision:.4f})")
print(f"Recall:     {recall_red:.4f} (Baseline: {recall:.4f})")
print("=====")

plt.figure(figsize=(5, 4))
sns.heatmap(confusion_matrix(y_test, y_pred_reduced), annot=True, fmt='d', cmap='Oranges',
            xticklabels=['Not Potable', 'Potable'], yticklabels=['Not Potable', 'Potable'])
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.title('Reduced Model Confusion Matrix')
plt.show()
```

Dropping lowest impact features: ['Organic_carbon', 'Trihalomethanes']

===== REDUCED MODEL PERFORMANCE =====
Accuracy: 0.6646 (Baseline: 0.6387)
Precision: 0.6286 (Baseline: 0.5525)
Recall: 0.3438 (Baseline: 0.3906)
=====

c:\Users\xyz\Github\BIAI\.venv\Lib\site-packages\sklearn\neural_network_multilayer_perceptron.py:785: ConvergenceWarning: Stochastic Optimizer: Maximum iterations (500) reached and the optimization hasn't converged yet.
warnings.warn(



Iteration 3: Hyperparameter Tuning via Comprehensive Grid Search

```
In [8]: # 1. Setup Grid Search Validation Architecture
warnings.filterwarnings("ignore", category=ConvergenceWarning)

param_grid = {
    'hidden_layer_sizes': [(64,), (128,), (64, 32), (128, 64)],
    'activation': ['relu', 'tanh'],
    'learning_rate_init': [0.001, 0.01],
    'max_iter': [800] # Increased to guarantee full algorithmic convergence
}

print("Executing Hyperparameter Optimization Grid Search...")
mlp_tune = MLPClassifier(solver='adam', random_state=42)

grid_search = GridSearchCV(
    estimator=mlp_tune,
    param_grid=param_grid,
    cv=3,
    scoring='accuracy',
    n_jobs=-1,
```



```
        verbose=1
    )
    grid_search.fit(X_train_reduced, y_train)

    print("\n===== GRID SEARCH RESULTS =====")
    print("Best Hyperparameters Found:")
    print(grid_search.best_params_)
    print(f"Best Cross-Validation Accuracy: {grid_search.best_score_:.4f}")
    print("=====")
```

Executing Hyperparameter Optimization Grid Search...
Fitting 3 folds for each of 16 candidates, totalling 48 fits

```
===== GRID SEARCH RESULTS =====
Best Hyperparameters Found:
{'activation': 'tanh', 'hidden_layer_sizes': (64,), 'learning_rate_init': 0.001, 'max_iter': 800}
Best Cross-Validation Accuracy: 0.6740
=====
```

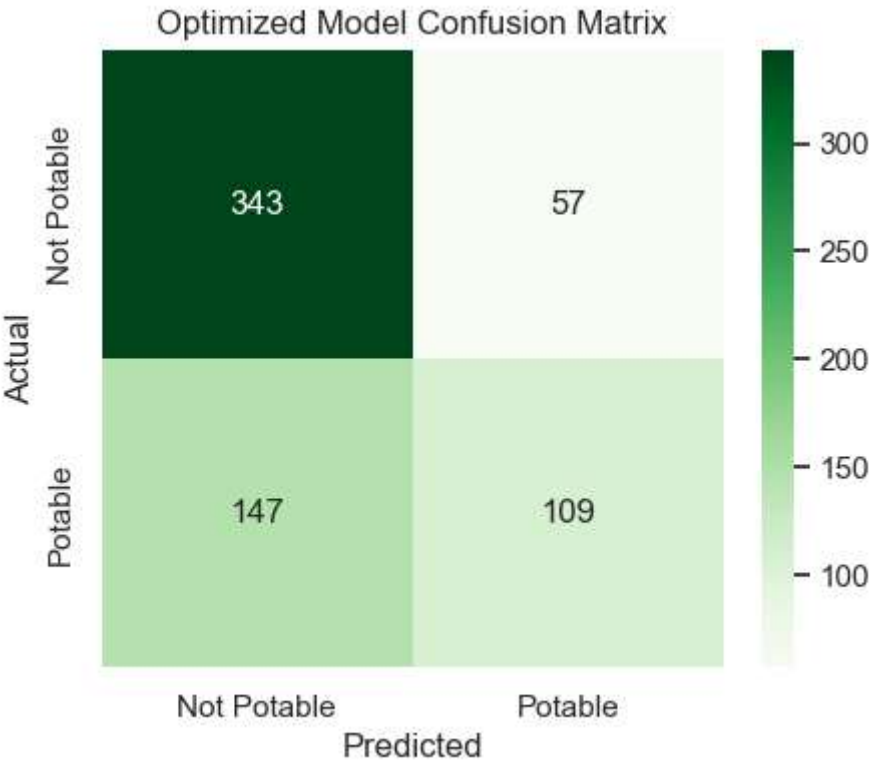
```
In [9]: # 2. Extract and Evaluate the Final Optimized Model
best_mlp = grid_search.best_estimator_
y_pred_opt = best_mlp.predict(X_test_reduced)

accuracy_opt = accuracy_score(y_test, y_pred_opt)
precision_opt = precision_score(y_test, y_pred_opt)
recall_opt = recall_score(y_test, y_pred_opt)

print("\n===== OPTIMIZED MODEL PERFORMANCE =====")
print(f"Accuracy:  {accuracy_opt:.4f} (Reduced Baseline: {accuracy_red:.4f})")
print(f"Precision: {precision_opt:.4f} (Reduced Baseline: {precision_red:.4f})")
print(f"Recall:    {recall_opt:.4f} (Reduced Baseline: {recall_red:.4f})")
print("=====")

plt.figure(figsize=(5, 4))
sns.heatmap(confusion_matrix(y_test, y_pred_opt), annot=True, fmt='d', cmap='Greens',
            xticklabels=['Not Potable', 'Potable'], yticklabels=['Not Potable', 'Potable'])
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.title('Optimized Model Confusion Matrix')
plt.show()
```

```
===== OPTIMIZED MODEL PERFORMANCE =====
Accuracy:  0.6890 (Reduced Baseline: 0.6646)
Precision: 0.6566 (Reduced Baseline: 0.6286)
Recall:    0.4258 (Reduced Baseline: 0.3438)
=====
```



Iteration 4: Project Summary & Performance Compilation

```
In [10]: # Compile metrics across all experimental cycles into a clean summary table
summary_data = {
    'Stage / Model Iteration': [
        '1. Baseline Model (All Features)',
        '2. Reduced Model (7 Features)',
        '3. Fully Optimized Model'
    ],
    'Accuracy': [accuracy, accuracy_red, accuracy_opt],
    'Precision': [precision, precision_red, precision_opt],
    'Recall': [recall, recall_red, recall_opt]
}
```

```
df_summary = pd.DataFrame(summary_data)
print("\n===== FINAL PROJECT EVALUATION SUMMARY =====")
display(df_summary.style.highlight_max(color='lightgreen', axis=0, subset=['Accuracy', 'Precision', 'Recall']))
```

===== FINAL PROJECT EVALUATION SUMMARY =====

	Stage / Model Iteration	Accuracy	Precision	Recall
0	1. Baseline Model (All Features)	0.638720	0.552486	0.390625
1	2. Reduced Model (7 Features)	0.664634	0.628571	0.343750
2	3. Fully Optimized Model	0.689024	0.656627	0.425781